AISC5020
Week 5
Probability Foundation

Presented by

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Chapter Outline

- **Probability Foundations**
- **Random variables, PMFs/PDFs, expectations, variance**
- **Joint, marginal, conditional distributions**
- **Covariance matrices**
- **Common distributions: Gaussian, Bernoulli, Multinomial**
- **Independence and conditional independence**

Probability Foundations

- The sample space Ω is the set of possible outcomes of an experiment. Points $\omega \in \Omega$ are called sample outcomes, realizations or elements.

  - If we toss a coin twice then $\Omega = \{HH, HT, TH, TT\}$.

- Subsets of Ω are called Events.

  - The event that the first toss is heads is $A = \{HH, HT\}$.

- Example. A coin is flipped 10 times. A possible outcome is $HHHTHHTTHT$, and the sample space is $\Omega = \{(\omega_1, \omega_2, \dots, \omega_{10}) : \omega_i \in \{H, T\}, 1 \leq i \leq 10\}$.

  - The event that the first head appears on the third toss is $E = \{(T, T, H, \omega_4, \dots, \omega_{10}) : \omega_i \in \{H, T\}, 4 \leq i \leq 10\}$.

  - The event that the first flip is Head is $A_1 = \{(H, \omega_2, \dots, \omega_{10}) : \omega_i \in \{H, T\}, 2 \leq i \leq 10\}$.

Probability Foundations

- Let $A, B \subseteq \Omega$ be events. $A \cup B$ is the event occurs if and only if at least one of A, B occurs. $A \cap B$ is the event occurs if and only both A, B occur.

- The complement of the event A, denoted A^c, is the event that A does not occur. Note that A and A^c are disjoint or mutually exclusive.

- For $1 \leq j \leq 10$, let $A_j = \{(\omega_1, \dots, \omega_{j-1}, H, \omega_{j+1}, \omega_{10}) : \omega_i \in \{H, T\}, 1 \leq i \leq 10, i \neq j\}$ be the event that the jth flip is Head.
  - The event that at least one flip was Heads is $B = \bigcup_{j=1}^{10} A_j$.
  - The event that all the flips were Heads is $C = \bigcap_{j=1}^{10} A_j$.

- $A_1, A_2, \dots$ are disjoint or are mutually exclusive if $A_i \cap A_j = \emptyset$ for $i \neq j$. A **partition** of Ω is a sequence of disjoint sets $A_1, A_2, \dots$ such that $\bigcup_{j=1}^{\infty} A_j = \Omega$.

Probability Foundations

- Given an event A, the indicator function of A is $I_A(\omega) = 1$ if $\omega \in A$ and $I_A(\omega) = 0$ if $\omega \notin A$.
- A sequence of events $A_1, A_2, \ldots$ is monotone increasing if $A_i \subseteq A_{i+1}$. Then $A = \bigcup_{i\in\mathbb{N}} A_i = \lim_{i\to\infty} A_i$ exists.
- A sequence of events $A_1, A_2, \ldots$ is monotone decreasing if $A_i \supseteq A_{i+1}$. Then $A = \bigcap_{i\in\mathbb{N}} A_i = \lim_{i\to\infty} A_i$ exists.
- Example. Let $\Omega = \mathbb{R}$ and let $A_i = \left[0, \frac{1}{i}\right)$ for $i = 1,2, \ldots$ Then $\bigcup_{i\in\mathbb{N}} A_i = [0,1)$ and $\bigcap_{i\in\mathbb{N}} A_i = \{0\}$. If $A_i = \left(0, \frac{1}{i}\right)$ then $\bigcup_{i\in\mathbb{N}} A_i = (0,1)$ and $\bigcap_{i\in\mathbb{N}} A_i = \emptyset$.

Probability Foundations

- A function $P(\cdot)$ that assigns a real number $P(A)$ to each event A is a probability distribution or a probability measure if it satisfies the following three axioms:
  - Axiom 1: $P(A) \geq 0$ for every A
  - Axiom 2: $P(\Omega) = 1$
  - Axiom 3: If $A_1, A_2, \ldots$ are disjoint, then $P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$.
- It is not always possible to assign a probability to every event A if the sample space is large, such as the whole real line. Instead, we assign probabilities to a limited class of set called a σ-field.

Not define here.

- $P(\emptyset) = 0$;
- $A \subset B \Longrightarrow P(A) \leq P(B)$;
- $0 \leq P(A) \leq 1$;
- $P(A^c) = 1 - P(A)$;
- $A \cap B = \emptyset \Longrightarrow P(A \cup B) = P(A) + P(B)$

Probability Foundations

- For any events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Example. Two-coin tosses. Let H_1 be the event that heads occurs on toss 1 and let H_2 be the event that heads occurs on toss 2. If all outcomes are equally likely,

$$P(H_1 \cup H_2) = P(H_1) + P(H_2) - P(H_1 \cap H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

- Suppose that $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$ is finite. If each outcome is equally likely, then $P(A) = \frac{|A|}{n}$ where $|A|$ denote the number of elements in A.
- Two events A and B are independent if $P(A \cap B) = P(A)P(B)$. A set of events $\{A_i : i \in I\}$ is independent if $P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j)$ for every finite subset J of I.
- Example. In tossing a fair die, let $A = \{2,4,6\}$ and let $B = \{1,2,3,4\}$. Then $A \cap B = \{2,4\}$, $P(A \cap B) = \frac{2}{6} = P(A)P(B) = \frac{1}{2} \times \frac{2}{3}$. So, A and B are independent.

Probability Foundations

- Disjoint events with positive probabilities are not independent since $P(A \cap B) = P(\emptyset) = 0$ but $P(A)P(B) > 0$.

- Example. Two people take turns trying to sink a basketball into a net. Person 1 succeeds with probability $\frac{1}{3}$ while person 2 succeeds with probability $\frac{1}{4}$. What is the probability that person 1 succeeds before person 2?

- Let E be the event of interest. Let A_j be the event that the first success is by person 1 and that it occurs on trial number j. Note that $A_1, A_2, \ldots$ are disjoint (mutually exclusive) and that $E = \bigcup_{j=1}^{\infty} A_j$. Hence, $P(E) = \sum_{j=1}^{\infty} P(A_j)$. Note that $P(A_1) = \frac{1}{3}$, $P(A_2) = \left(\frac{2}{3} \times \frac{3}{4}\right)\frac{1}{3} = \frac{1}{2} \cdot \frac{1}{3}$, etc. Thus, $P(A_j) = \left(\frac{2}{3} \times \frac{3}{4}\right)^{j-1}\frac{1}{3} = \left(\frac{1}{2}\right)^{j-1}\frac{1}{3}$. Consequently, $P(E) = \sum_{j=1}^{\infty}\left(\frac{1}{2}\right)^{j-1}\frac{1}{3} = \frac{1}{3} \times \frac{1}{1-\frac{1}{2}} = \frac{2}{3}$.

Probability Foundations

- If $P(B) > 0$ then the conditional probability of A given B is $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
  - Since $P(\cdot\,|B)$ is a probability that satisfies the three axioms, $P(\Omega|B) = 1$.
  - If $A_1, A_2, \ldots$ are disjoint then $P\left(\bigcup_{j=1}^{\infty} A_j \,\middle|\, B\right) = \sum_{j=1}^{\infty} P(A_j|B)$.
  - In general, $P(A|B \cup C) \neq P(A|B) + P(A|C)$.
  - In general, $P(A|B) \neq P(B|A)$.

Probability Foundations

- Example. A medical test for a disease D has outcomes $+$ and $-$. The probabilities are

	D	D^c
$+$	0.009	0.099
$-$	0.001	0.891

$$P(+|D) = \frac{P(+\cap D)}{P(D)} = \frac{0.009}{0.009 + 0.001} = 0.9.$$

$$P(-|D^c) = \frac{P(-\cap D^c)}{P(D^c)} = \frac{0.891}{0.099 + 0.891} \approx 0.9.$$

- Suppose you go for a test and get a positive, what is the probability you have the disease?

$$P(D|+) = \frac{P(+\cap D)}{P(+)} = \frac{0.009}{0.009 + 0.099} \approx 0.08.$$

Probability Foundations

- If A and B are independent events, then $P(A|B) = P(A)$. Also, for any pair of events A and B,

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A).$$

- Example. Draw two cards from a deck, without replacement. Let A be the event that the first draw is the Ace of Clubs and let B be the event that the second draw is the Queen of Diamonds. Then $P(A \cap B) = P(A)P(B|A) = \frac{1}{52} \times \frac{1}{51}$.

- The Law of Total Probability. Let $A_1, A_2, \ldots, A_k$ be a partition of Ω. Then, for any event B,

$$P(B) = P(B \cap \Omega) = P\left(B \cap \left(\bigcup_{j=1}^{k} A_k\right)\right) = P\left(\bigcup_{j=1}^{k} B \cap A_k\right) = \sum_{j=1}^{k} P(B \cap A_k) = \sum_{j=1}^{k} P(B|A_k)P(A_k)$$

- since $B \cap A_1, B \cap A_2, \ldots, B \cap A_k$ are disjoint.

Probability Foundations

- Bayes' Theorem. Let $A_1, A_2, \ldots, A_k$ be a partition of Ω such that $P(A_i) > 0$ for each i. If $P(B) > 0$ then, for each $i = 1,2, \ldots, k$,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{k} P(B|A_j)P(A_j)}.$$

- $P(A_i)$ is the prior probability of A_i and $P(A_i|B)$ is the posterior probability of A_i.

- Example. I divide my email into three categories: $A_1 =$ "spam", $A_2 =$ "low priority", $A_3 =$ "high priority". From previous experience, I find that $P(A_1) = 0.7$, $P(A_2) = 0.2$ and $P(A_3) = 0.1$. Let B be the event that the email contains the word "free". From previous experience, $P(B|A_1) = 0.9$, $P(B|A_2) = 0.01$ and $P(B|A_3) = 0.01$. I receive an email with the word "free". What is the probability that it is spam?

$$P(A_1|B) = \frac{0.9 \times 0.7}{0.9 \times 0.7 + 0.01 \times 0.2 + 0.01 \times 0.1} = 0.995$$

Random Variables

- A random variable $X\colon\ \Omega \to \mathbb{R}$ assigns a real number $X(\omega)$ to each outcome ω.
- Example. Flip a coin ten times. Let $X(\omega)$ be the number of heads in the sequence ω. For example, if $\omega = HHTHHTHHTT$, then $X(\omega) = 6$.
- Example. Let $\Omega = \{(x, y)\colon\ x^2 + y^2 \leq 1\}$. Consider drawing a point at random from Ω. Some examples of random variables are $X(\omega) = x$, $Y(\omega) = y$, $Z(\omega) = x + y$ and $W(\omega) = \sqrt{x^2 + y^2}$.
- Given a random variable X and a subset A of the real line, define $X^{-1}(A) = \{\omega \in \Omega\colon X(\omega) \in A\}$ and let $P(X \in A) = P(X^{-1}(A) \subseteq \Omega) = P(\{\omega \in \Omega\colon X(\omega) \in A\})$. $P(X = x) = P(\{\omega \in \Omega\colon X(\omega) = x\})$.
- Example. Flip a coin twice and let X be the number of heads.

$\boldsymbol{\omega}$	$\boldsymbol{P(\{\omega\})}$	$\boldsymbol{X(\omega)}$
TT	0.25	0
TH	0.25	1
HT	0.25	1
HH	0.25	2

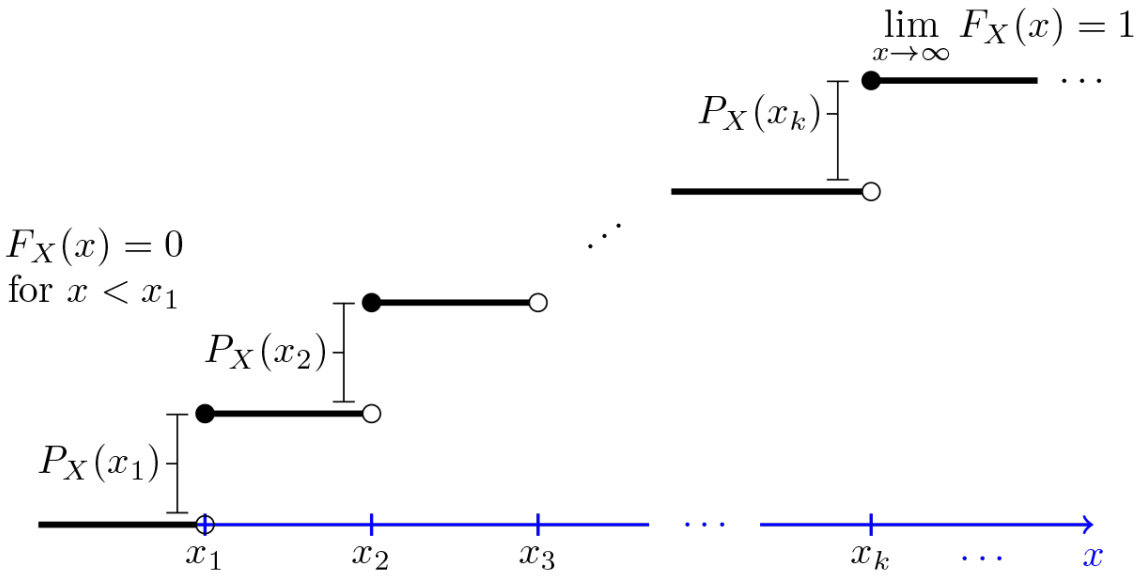
$\boldsymbol{x}$	$\boldsymbol{P(X = x)}$
0	0.25
1	0.5
2	0.25

Distribution Functions and Probability Functions

- The cumulative distribution function, or CDF, is the function $F_X: \mathbb{R} \to [0,1]$ defined by

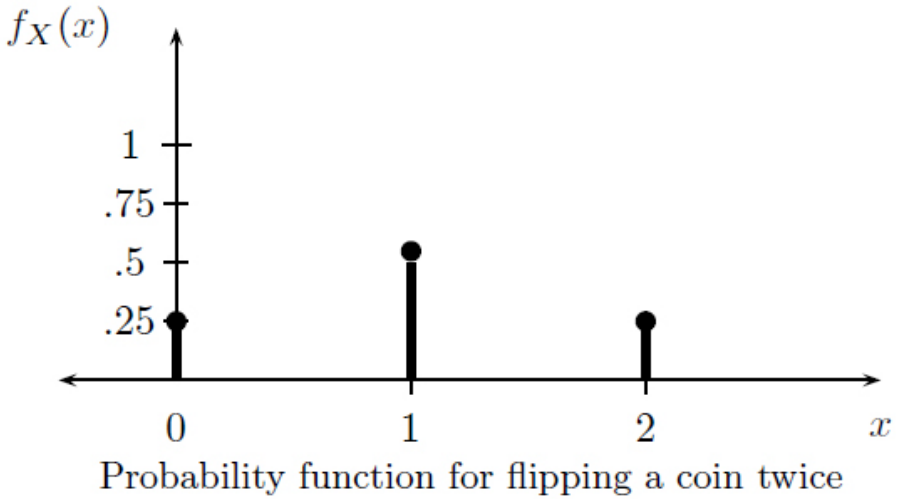
$$F_X(x) = P(X \leq x).$$

- Let X have CDF F and let Y have CDF G. If $F(x) = G(x)$ for all x, then $P(X \in A) = P(Y \in A)$ for all A.
- Properties
  - F is non-decreasing.
  - $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$
  - F is right-continuous: $F(x) = F(x^+)$ for all x



Distribution Functions and Probability Functions

- X is discrete if it takes countably many values $\{x_1, x_2, \dots\}$. We define the probability mass function for X by $f_X(x) = P(X = x)$.
- Properties
  - $f_X(x) \geq 0$ for all $x \in \mathbb{R}$
  - $\sum_i f_X(x_i) = 1$.
- The CDF of X is $F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$.

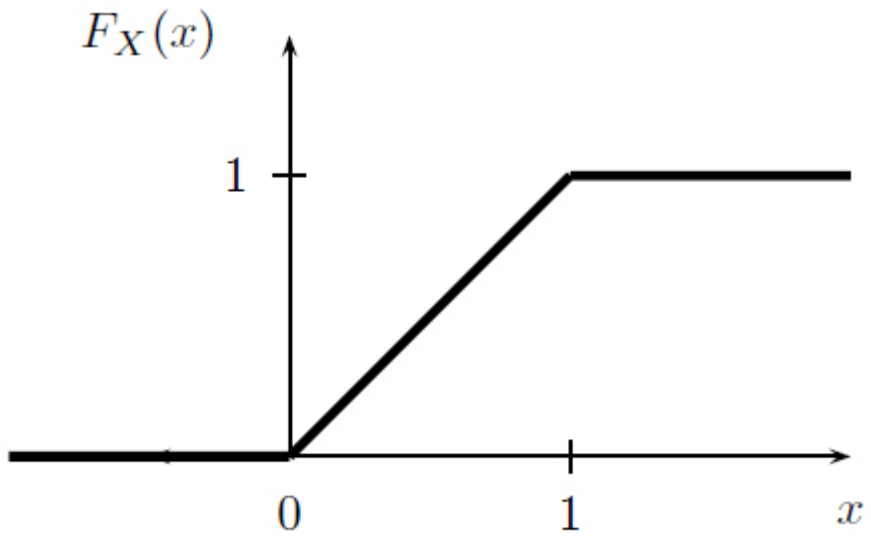


Probability function for flipping a coin twice

- A random variable X is continuous if there exists a function f_X such that $f_X(x) \geq 0$ for all $x \in \mathbb{R}$, $\int_{-\infty}^{\infty} f_X(x)dx = 1$ and for every $a \leq b$, $P(a < X < b) = \int_a^b f_X(x)dx$.
- The function f_X is called the probability density function (pdf). We have $F_X(x) = \int_{-\infty}^{x} f_X(t)dt$ and $f_X(x) = \frac{dF_X(x)}{dx}$ at all points x at which F_X is differentiable.

Distribution Functions and Probability Functions

- Example. Suppose that X has pdf $f_X(x) = 1$ for $0 \leq x \leq 1$ and $f_X(x) = 0$ otherwise. Clearly, $f_X(x) \geq 0$ and $\int_0^1 f_X(x)dx = 1$. The CDF is $F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x \leq 1. \\ 1 & x > 1 \end{cases}$



CDF for Uniform (0,1).

Distribution Functions and Probability Functions

- If X is continuous, then $P(X = x) = 0$ for every x.
- A pdf can be bigger than 1. For example, $f_X(x) = 5$ for all $x \in \left[0, \frac{1}{5}\right]$ and $f_X(x) = 0$ otherwise.
- A pdf can be unbounded. For example, $f_X(x) = \frac{2}{3}x^{-\frac{1}{3}}$ for $0 < x < 1$ and $f_X(x) = 0$ otherwise.
- Let X be a random variable with CDF F. The inverse CDF or quantile function is defined by $F^{-1}(q) = \inf\{x: F(x) > q\}$ for $q \in [0,1]$. If F is strictly increasing and continuous then $F^{-1}(q)$ is the unique real number x such that $F(x) = q$.
  - $F^{-1}(0.25)$ is the first quartile, $F^{-1}(0.5)$ is the median and $F^{-1}(0.75)$ is the third quartile.
- Two random variables X and Y are equal in distribution – written $X \overset{d}{\Leftrightarrow} Y$ if $F_X(x) = F_Y(x)$ for all x. But X and Y are not necessarily equal. For example, $P(X = 1) = P(X = -1) = 0.5$. $Y = -X \overset{d}{\Leftrightarrow} X$.

Important Discrete Random Variables

- The Point Mass Distribution. X has a point mass distribution at a written $X \sim \delta_a$ if $P(X = a) = 1$ in which case $f_X(x) = 1$ for $x = a$ and $f_X(x) = 0$ for $x \neq a$ and $F_X(x) = \begin{cases} 0 & x < a \\ 1 & x \geq a \end{cases}$.
- The Discrete Uniform Distribution. Let $k > 1$ be a given integer. Suppose that X has probability mass function given by $f_X(x) = \begin{cases} \frac{1}{k} & \text{for } x = 1,2,\dots,k \\ 0 & \text{otherwise} \end{cases}$.
- The Bernoulli Distribution. Let X represent a binary coin flip. Suppose $P(X = 1) = p$ and $P(X = 0) = 1 - p$ for some $p \in [0,1]$. We say that $X \sim Bernoulli(p)$ and has pmf $f_X(x) = p^x(1-p)^{1-x}$ for $x \in \{0,1\}$.
- The Binomial Distribution. Suppose we have a coin which falls heads up with probability p for some $0 \leq p \leq 1$. Flip the coin n times and let X be the number of heads. Assume that the tosses are independent. Then $f_X(x) = \binom{n}{x} p^x(1-p)^{n-x}$ for $x = 0,1,2,\dots,n$. We write $X \sim Binomial(n,p)$.
  - If $X_1 \sim Binomial(n_1,p)$ and $X_2 \sim Binomial(n_2,p)$ and X_1 and X_2 are independent,

$$X_1 + X_2 \sim Binomial(n_1 + n_2, p).$$

Important Discrete Random Variables

- The Geometric Distribution. X has a geometric distribution with parameter $p \in (0,1)$, written $X \sim G_1(p)$ if $f_X(k) = p(1-p)^{k-1}, k = 1,2, \ldots$ Think of X as the number of flips needed until the first head when flipping a coin.
- The Poisson Distribution. X has a Poisson distribution with parameter λ, written $X \sim Poisson(\lambda)$ if $f_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}, k = 0,1,2, \ldots$ If $X_1 \sim Poisson(\lambda_1)$, $X_2 \sim Poisson(\lambda_2)$ and X_1 and X_2 are independent, then $X_1 + X_2 \sim Poisson(\lambda_1 + \lambda_2)$.

Important Continuous Random Variables

- The Uniform Distribution. X has a uniform (a,b) distribution if $f_X(x) = \frac{1}{b-a} I\big(x \in (a,b)\big), a < b.$
- The Normal Distribution. X has a normal distribution with parameters μ and σ^2, denoted by $X \sim n(\mu, \sigma^2)$, if

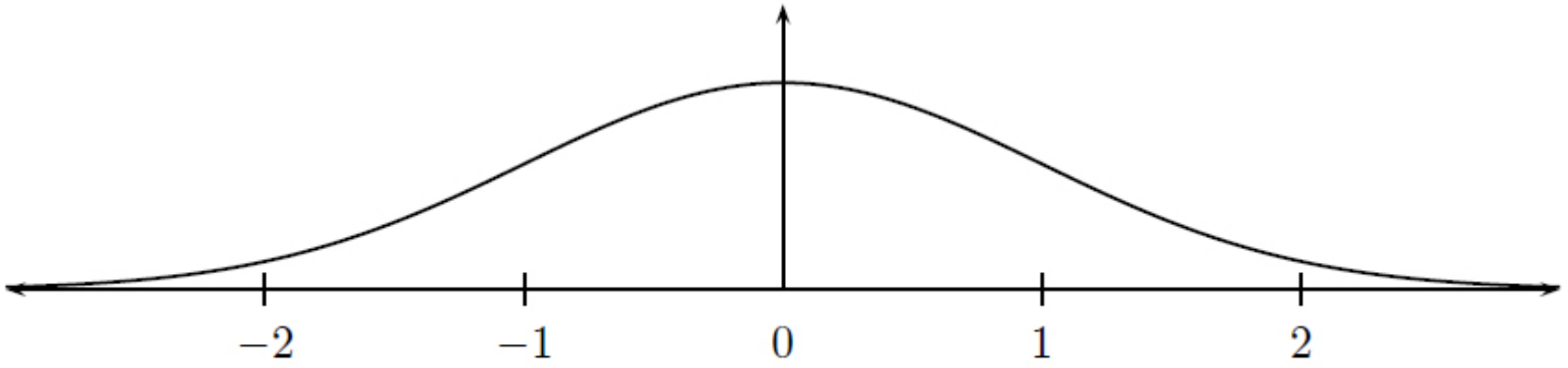
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$. X has a standard normal distribution if $\mu = 0$ and $\sigma = 1$. The pdf and CDF of a standard normal are denoted by $\phi(z)$ and $\Phi(z)$.

- If $X \sim n(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim n(0,1)$.
- If $Z \sim n(0,1)$, then $X = \mu + \sigma Z \sim n(\mu, \sigma^2)$.
- If $X_i \sim n\left(\mu_i, \sigma_i^2\right), i = 1,2,\dots,n$ are independent, then $\sum_{i=1}^{n} X_i \sim n\left(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2\right)$.
- If $X \sim n(\mu, \sigma^2)$, then

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Important Continuous Random Variables



Density of a standard Normal.

- Example. Suppose that $X \sim n(3,5)$. Find $P(X > 1)$.

$$P(X > 1) = 1 - P(X < 1) = 1 - P\left(Z < \frac{1-3}{\sqrt{5}}\right) = 1 - \Phi(-0.8944) = 0.81.$$

- Example. Find $q = F_X^{-1}(0.2) \Longrightarrow F_X(q) = 0.2$.

$$0.2 = P(X < q) = P\left(Z < \frac{q-\mu}{\sigma}\right) = \Phi\left(\frac{q-\mu}{\sigma}\right)$$

Since $\Phi(-0.8416) = 0.2$, $\frac{q-3}{\sqrt{5}} = -0.8416 \Longrightarrow q = 3 - 0.8416\sqrt{5} = 1.1181$.

Important Continuous Random Variables

- Exponential Distribution. X has an exponential distribution with parameter β, denoted by $X \sim Exp(\beta)$ if $f_X(x) = \beta e^{-\beta x}, x > 0, \beta > 0$. The CDF is $F_X(x) = 1 - e^{-\beta x}, x > 0$.

- Gamma Distribution. For $\alpha, \beta > 0$, $X \sim Gamma(\alpha, \beta)$ if $f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x > 0$.

  The exponential distribution is a $Gamma(1, \beta)$ distribution. If $X_i \sim Gamma(\alpha_i, \beta)$ are independent, then $\sum_{i=1}^{n} X_i \sim Gamma\left(\sum_{i=1}^{n} \alpha_i, \beta\right)$.

- The Beta Distribution. X has a Beta distribution with parameters $\alpha > 0$ and $\beta > 0$, denoted by $X \sim Beta(\alpha, \beta)$, if $f_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, 0 < x < 1$.

- t and Cauchy Distribution. X has a t distribution with ν degrees of freedom, written $X \sim t_\nu$, if $f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$. Note that $\lim_{\nu \to \infty} t_\nu \overset{d}{\Leftrightarrow} Z \sim n(0,1)$. The t distribution is similar to a Normal but it has thicker tails.

Important Continuous Random Variables

- The special case t_1 is called the Cauchy distribution. The pdf is $f_X(x) = \frac{1}{\pi(1+x^2)}, x \in \mathbb{R}$. Mean and Variance of the Cauchy distribution is undefined.
- The χ^2 distribution. For a positive integer p, $X \sim Gamma\left(\frac{p}{2}, \frac{1}{2}\right)$ is called the χ^2 distribution with p degrees of freedom, written $X \sim \chi^2_p$.
  - If $Z_1, Z_2, \ldots, Z_p$ are independent standard normal random variables, then $\sum_{j=1}^{n} Z_j^2 \sim \chi^2_p$.

Bivariate Distributions

- Given a pair of discrete random variables, the joint pmf is $f_{X,Y}(x,y) = P(X = x, Y = y)$.
- Example. A bivariate distribution for X and Y is

	$Y = 0$	$Y = 1$	
$X = 0$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$
$X = 1$	$\frac{2}{9}$	$\frac{4}{9}$	$\frac{2}{3}$
	$\frac{1}{3}$	$\frac{2}{3}$	1

- Thus, $f(1,1) = P(X = 1, Y = 1) = \frac{4}{9}$.
- A pdf for continuous bivariate random variables (X, Y) is $f_{X,Y}(x,y)$ if it satisfies
  - $f_{X,Y}(x,y) \geq 0$ for all (x,y)
  - $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy = 1$
  - for any set $A \subset \mathbb{R} \times \mathbb{R}$, $P\big((X,Y) \in A\big) = \int\int_A f_{X,Y}(x,y)dxdy$
- In the discrete or continuous case, we define the joint CDF as $F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$.

Bivariate Distributions

- Example. Let (X, Y) have joint pdf $f_{X,Y}(x, y) = \begin{cases} cx^2y & \text{if } x^2 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$. Find the value of c and compute $P(X \ge Y)$.
- $x^2 \le 1 \Longrightarrow -1 \le x \le 1$.

$$1 = \int_{-1}^{1} \int_{x^2}^{1} cx^2 y dy\, dx = \int_{-1}^{1} cx^2 \frac{1 - x^4}{2} dx = \frac{c}{2} \int_{-1}^{1} (x^2 - x^6) dx = \frac{4c}{21} \Longrightarrow c = \frac{21}{4}.$$

$$P(X \ge Y) = \frac{21}{4} \int_{0}^{1} \int_{x^2}^{x} x^2 y dy\, dx = \frac{21}{4} \int_{0}^{1} x^2 \left(\frac{x^2 - x^4}{2} \right) dx = \frac{3}{20}.$$

Bivariate Distributions

- If (X, Y) have joint pmf $f_{X,Y}$ then the marginal pmf for X is $f_X(x) = \sum_y f_{X,Y}(x, y)$ and the marginal pmf for Y is $f_Y(y) = \sum_x f_{X,Y}(x, y)$.
- For example, $f_X(0) = \frac{1}{3}$ and $f_X(1) = \frac{2}{3}$.

	$Y = 0$	$Y = 1$	
$X = 0$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$
$X = 1$	$\frac{2}{9}$	$\frac{4}{9}$	$\frac{2}{3}$
	$\frac{1}{3}$	$\frac{2}{3}$	1

- For continuous random variables, the marginal pdfs are $f_X(x) = \int f_{X,Y}(x, y)dy$ and $f_Y(y) = \int f_{X,Y}(x, y)dx$.
- Suppose that $f_{X,Y}(x, y) = \begin{cases} \frac{21}{4}x^2y & \text{if } x^2 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$. Then, $f_X(x) = \int_{x^2}^1 \frac{21}{4}x^2y dy = \frac{21}{8}x^2(1 - x^4)$ for $|x| < 1$ and $f_X(x) = 0$ otherwise.

Bivariate Distributions

- Let X and Y have joint pdf or pmf $f_{X,Y}$. Then, X and Y are independent if and only if $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all values x and y.
- Example. Suppose that X and Y are independent and both have the same density $f_X(x) = \begin{cases} 2x \text{ if } 0 \le x \le 1 \\ 0 \quad \text{otherwise} \end{cases}$. Find $P(X + Y \le 1)$.
- Using independence, the joint pdf is $f_{X,Y}(x,y) = f_X(x)f_Y(y) = \begin{cases} 4xy \text{ if } 0 \le x \le 1, 0 \le y \le 1 \\ 0 \quad \text{otherwise} \end{cases}$.

$$P(X + Y \le 1) = 4\int_0^1 x \int_0^{1-x} y dy\, dx = \frac{1}{6}.$$

- Suppose that the range of X and Y is a rectangle. If $f_{X,Y}(x,y) = g(x)h(y)$ for some functions g and h then X and Y are independent.
- Example. Let X and Y have joint pdf $f_{X,Y}(x,y) = \begin{cases} 2e^{-(x+2y)} \text{ if } x > 0, y > 0 \\ 0 \quad \text{otherwise} \end{cases}$. The range of X and Y is a rectangle $(0,\infty) \times (0,\infty)$. Since $f_{X,Y}(x,y) = 2e^{-x}e^{-2y}$, X and Y are independent.

Conditional Distributions

- The conditional pmf or pdf is $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ if $f_Y(y) > 0$. In the continuous case,

$$P(X \in A|Y = y) = \int_A f_{X|Y}(x|y)dx\,.$$

- Example. Suppose that $X \sim \text{Uniform}(0,1)$ and $Y|X = x \sim \text{Uniform}(x,1)$. What is the marginal distribution of Y.
- The marginal distribution for Y is

$$f_Y(y) = \int_0^1 f_{Y|X=x}(y|x)\, f_X(x)dx = \int_0^1 \frac{1}{1-x} I(x < y < 1)dx = \int_0^y \frac{1}{1-x}dx$$
$$= -\ln(1-y)\,, 0 < y < 1$$

Conditional Distributions

- Example. Consider the density $f_{X,Y}(x,y) = \begin{cases} \frac{21}{4}x^2y & \text{if } x^2 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$. In p.26, we showed that $f_X(x) = \frac{21}{8}x^2(1-x^4)I(-1 < x < 1)$. For $x^2 \le y \le 1$,

$$f_{Y|X=x}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{21}{4}x^2y}{\frac{21}{8}x^2(1-x^4)} = \frac{2y}{1-x^4}.$$

- To compute $P\left(Y \ge \frac{3}{4} \middle| X = \frac{1}{2}\right)$.

$$P\left(Y \ge \frac{3}{4} \middle| X = \frac{1}{2}\right) = \int_{\frac{3}{4}}^{1} f_{Y\middle|X=\frac{1}{2}}\left(y \middle| \frac{1}{2}\right) dy = \int_{\frac{3}{4}}^{1} \frac{32}{15} y dy = \frac{7}{15}.$$

Multivariate Distributions and IID samples

- If $X_1, X_2, \ldots, X_n$ are independent and each has the same marginal distribution with CDF F, we say that $X_1, X_2, \ldots, X_n$ are IID (independently and identically distributed) and write $X_1, X_2, \ldots, X_n \sim F$. If F has pdf f, we also write $X_1, X_2, \ldots, X_n \sim f$. We say $X_1, X_2, \ldots, X_n$ a random sample of size n from F.
- Multinomial. Consider drawing a ball from an urn which has balls with k different colors labeled "color 1, color 2, ..., color k". Let $p = (p_1, p_2, \ldots, p_k)$ where $p_j \geq 0$ and $\sum_{j=1}^{k} p_j = 1$ and suppose that p_j is the probability of drawing a ball of color j. Draw n balls independently, with replacement, and let $X = (X_1, X_2, \ldots, X_k)$ where X_j is the number of balls of color j. It is required that $\sum_{j=1}^{k} X_j = n$. We say that X has a Multinomial(n, p) distribution. The joint pmf is

$$f_{(X_1, X_2, \ldots, X_k)}(x_1, x_2, \ldots, x_k) = \frac{n!}{x_1!\, x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

- The marginal distribution of X_j is Binomial(n, p_j).

Multivariate Distributions and IID samples

- Multivariate Normal. Let $Z = \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_k \end{pmatrix}$, where $Z_1, Z_2, \dots, Z_k \sim n(0,1)$ are independent. The pdf of Z is

$$f_{(Z_1,Z_2,\dots,Z_k)}\big(z = (z_1, z_2, \dots, z_k)\big) = \prod_{i=1}^{k} f_Z(z_i) = \prod_{i=1}^{k} \frac{1}{\sqrt{2\pi}} e^{-\frac{z_i^2}{2}} = (2\pi)^{-\frac{k}{2}} e^{-\frac{1}{2}\sum_{j=1}^{k} z_j^2}$$

$$= (2\pi)^{-\frac{k}{2}} e^{-\frac{1}{2}z'z}.$$

  - We write that $Z \sim n(0, I_k)$, where 0 represents a vector of k zeros and I_k is the $k \times k$ identity matrix.

- X has a multivariate normal distribution, denoted by $X \sim n(\mu, \Sigma)$, if it has pdf

$$f_{X=(X_1,X_2,\dots,X_k)}\big(x = (x_1, x_2, \dots, x_k)\big) = (2\pi)^{-\frac{k}{2}} |\Sigma|^{-\frac{1}{2}} e^{-\frac{1}{2}(x-\mu)'\Sigma^{-1}(x-\mu)}$$

  where $|\Sigma|$ is the determinant of Σ, μ is a $k \times 1$ vector and Σ is a $k \times k$ symmetric, positive definite matrix. Setting $\mu = 0$ and $\Sigma = I_k$ give the standard normal.

Multivariate Distributions and IID samples

- Properties of symmetric and positive definite matrix Σ
  - Existence of a matrix $\Sigma^{\frac{1}{2}}$
  - $\Sigma^{\frac{1}{2}}$ is symmetric
  - $\Sigma = \Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}}$
  - $\Sigma^{\frac{1}{2}}\Sigma^{-\frac{1}{2}} = \Sigma^{-\frac{1}{2}}\Sigma^{\frac{1}{2}} = I$
- If $Z \sim n(0, I_k)$ and $X = \mu + \Sigma^{\frac{1}{2}}Z$ then $X \sim n(\mu, \Sigma)$.
- If $X \sim n(\mu, \Sigma)$, then $\Sigma^{-\frac{1}{2}}(X - \mu) \sim n(0, I_k)$.
- $\left(\Sigma^{-\frac{1}{2}}(X - \mu)\right)' \Sigma^{-\frac{1}{2}}(X - \mu) \sim \chi^2_k \Longrightarrow (X - \mu)'\Sigma^{-1}(X - \mu) \sim \chi^2_k$

Multivariate Distributions and IID samples

- Suppose we partition a random Normal vector $X = (X_a, X_b)$ and partition similarly $\mu = (\mu_a, \mu_b)$ and $\Sigma = \begin{pmatrix} \Sigma_{a,a} & \Sigma_{a,b} \\ \Sigma_{b,a} & \Sigma_{b,b} \end{pmatrix}$.
- Let $X \sim n(\mu, \Sigma)$. Then
  - If A is a $m \times k$ matrix, then $AX \sim n(A\mu, A\Sigma A')$.
  - The marginal distribution of X_a is $X_a \sim n(\mu_a, \Sigma_{a,a})$. Note that $X_a = (I_a \quad 0)X$ and a linear combination of normal vectors is normal.
  - $X_b | X_a = x_a \sim n(\mu_b + \Sigma_{b,a}\Sigma_{a,a}^{-1}(x_a - \mu_a), \Sigma_{b,b} - \Sigma_{b,a}\Sigma_{a,a}^{-1}\Sigma_{a,b})$

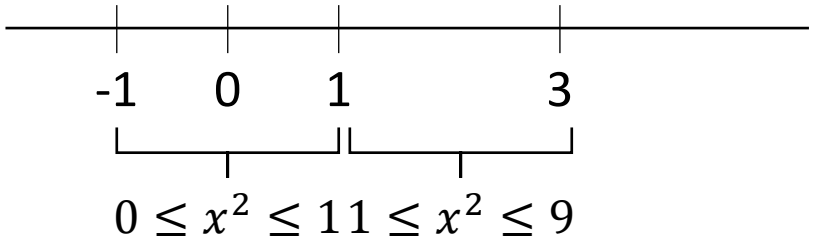
Consider $A = \begin{pmatrix} I_a & 0 \\ -\Sigma_{b,a}\Sigma_{a,a}^{-1} & I_b \end{pmatrix}$. Then, $AX = \begin{pmatrix} X_a \\ X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a \end{pmatrix} \sim n\left(\begin{pmatrix} \mu_a \\ \mu_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}\mu_a \end{pmatrix}, \begin{pmatrix} \Sigma_{a,a} & 0 \\ 0 & \Sigma_{b,b} - \Sigma_{b,a}\Sigma_{a,a}^{-1}\Sigma_{a,b} \end{pmatrix} \right)$

Thus, X_a and $X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a$ are independent and $X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a \sim n(\mu_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}\mu_a, \Sigma_{b,b} - \Sigma_{b,a}\Sigma_{a,a}^{-1}\Sigma_{a,b})$.

Since $X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a | X_a = x_a \overset{d}{\Leftrightarrow} X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a, X_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}X_a | X_a = x_a \sim n(\mu_b - \Sigma_{b,a}\Sigma_{a,a}^{-1}\mu_a, \Sigma_{b,b} - \Sigma_{b,a}\Sigma_{a,a}^{-1}\Sigma_{a,b})$
$\Rightarrow X_b | X_a = x_a \sim n(\mu_b + \Sigma_{b,a}\Sigma_{a,a}^{-1}(x_a - \mu_a), \Sigma_{b,b} - \Sigma_{b,a}\Sigma_{a,a}^{-1}\Sigma_{a,b})$

Transformations of Random Variables

- Example. Suppose that $P(X=-1)=P(X=1)=0.25$ and $P(X=0)=0.5$. Let $Y=X^2$. Then
  - $P(Y=0)=P(X=0)=0.5$
  - $P(Y=1)=P(X=1)+P(X=-1)=0.5$
- Example. Let $X\sim$Uniform$(-1,3)$. Find the pdf of $Y=X^2$.
  - Range of Y is $(0,9)$.
  - Case (i) $0<y<1$



$$F_Y(y)=P(Y<y)=P(-\sqrt{y}<X<\sqrt{y})=\int_{-\sqrt{y}}^{\sqrt{y}}\frac{1}{4}dx=\frac{\sqrt{y}}{2}\Rightarrow f_Y(y)=\frac{1}{4\sqrt{y}}$$

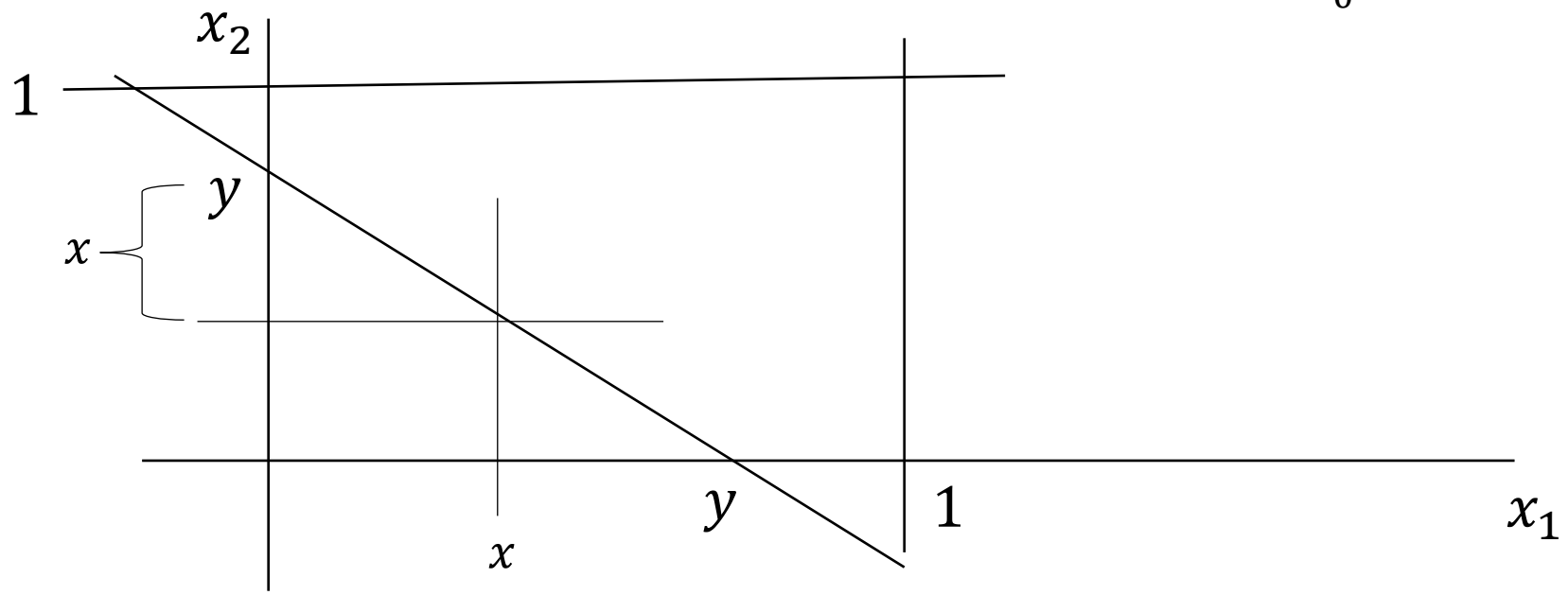
  - Case (ii) $1<y<9$

$$F_Y(y)=P(Y<y)=P(\underbrace{-1<X<}_{X\in(-1,3)}\sqrt{y})=\int_{-1}^{\sqrt{y}}\frac{1}{4}dx=\frac{1+\sqrt{y}}{4}\Rightarrow f_Y(y)=\frac{1}{8\sqrt{y}}$$

Transformations of Random Variables

- Example. Let $X_1, X_2 \sim \text{Uniform}(0,1)$ be independent. Find the pdf of $Y = X_1 + X_2$.
- Let $0 < y < 2$. Two cases: $0 < y < 1$ and $1 < y < 2$.
- Case 1: $0 < y < 1, 0 < x_1 < y, x_1 + x_2 < y \Longrightarrow 0 < x_2 < y - x_1$

$$P(X_1 + X_2 < y) = \int_0^y P(X_2 < y - x_1) f_{X_1}(x_1) dx_1 = \int_0^y (y - x)\, dx = -\frac{(y-x)^2}{2}\Big|_0^y = \frac{y^2}{2} \Longrightarrow f_{X_1+X_2}(y) = y$$



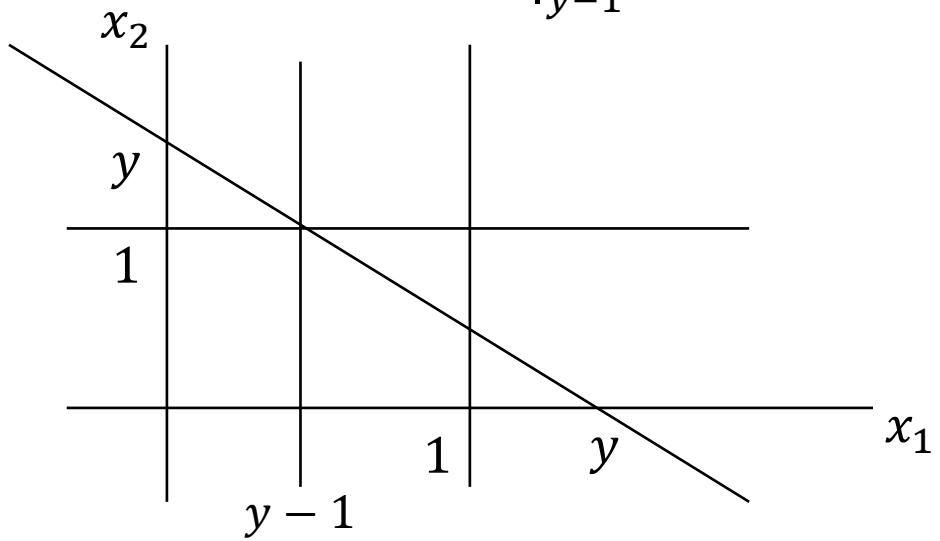
Transformations of Random Variables

- Case 2: $1 < y < 2$
  - $0 < x_1 < y-1, x_1 + x_2 < y \Longrightarrow x_2 < 1$
  - $y-1 < x_1 < 1, x_1 + x_2 < y \Longrightarrow 0 < x_2 < y - x_1$

$$P(X_1 + X_2 < y) = \int_0^{y-1} P(X_2 < 1) f_{X_1}(x_1) dx_1 + \int_{y-1}^{1} P(X_2 < y - x_1) f_{X_1}(x_1) dx_1$$

$$= \int_0^{y-1} 1\, dx + \int_{y-1}^{1} (y-x)\, dx = (y-1) - \frac{(y-x)^2}{2}\Big|_{y-1}^{1} = y - 1 + \frac{1}{2} - \frac{(y-1)^2}{2} = 1 - \frac{(y-2)^2}{2}$$

$$\Longrightarrow f_{X_1+X_2}(y) = 2 - y$$



Transformations of Random Variables

- Let X have pdf $f_X(x)$ and let $Y = g(X)$, where g is a montone function. Suppose that $f_X(x)$ is a continuous function on $\mathcal{X} = \{x \in \mathbb{R}: f_X(x) > 0\}$ and that $g^{-1}(y)$ has a continuous derivative on $\mathcal{Y} = \{y: y = g(x) \text{ for some } x \in \mathcal{X}\}$. Then the pdf of Y is

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\frac{d}{dy}g^{-1}(y)\right|, y \in \mathcal{Y}$$

- Example. Let $f_X(x)$ be the gamma pdf $f_X(x) = \frac{1}{(n-1)!\beta^n}x^{n-1}e^{-\frac{x}{\beta}}, x > 0, \beta > 0$. Suppose $g(X) = \frac{1}{X}$.
- Let $y = g(x) = \frac{1}{x} \Longrightarrow x = \frac{1}{y}. \frac{dx}{dy} = -\frac{1}{y^2}$. Thus, for $y > 0$,

$$f_Y(y) = \frac{1}{(n-1)!\,\beta^n}\left(\frac{1}{y}\right)^{n-1} e^{-\frac{1}{\beta y}}\left|\frac{dx}{dy}\right| = \frac{1}{(n-1)!\,\beta^n}\left(\frac{1}{y}\right)^{n-1} e^{-\frac{1}{\beta y}}\frac{1}{y^2} = \frac{1}{(n-1)!\,\beta^n}\left(\frac{1}{y}\right)^{n+1} e^{-\frac{1}{\beta y}}$$

Expectation of Random Variables

- The expected value, or mean, or first moment, of X is defined to be
  - $E(X) = \int x dF(x) = \sum_x x P(X = x)$ if X is discrete.
  - $E(X) = \int x dF(x) = \int x f_X(x) dx$ if X is continuous.
  - The expectation is a one-number summary of the distribution. Think of $E(X)$ as the average $\frac{1}{n}\sum_{j=1}^{n} X_j$ of a large number of IID draws $X_1, X_2, \ldots, X_n$.
- To ensure that $E(X)$ is well-defined, $E(X)$ exists if $\int |x| f_X(x) dx < \infty$.
- Example. Let $X \sim Bernoulli(p)$. Then $E(X) = 0 \times (1-p) + 1 \times p = p$.
- Example. Flip a fair coin twice. Let X be the number of heads. $E(X) = 0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} = 1$.
- Example. Let $X \sim$ Uniform$(-1,3)$. Then, $E(X) = \int_{-1}^{3} \frac{x}{4} dx = \frac{8}{8} = 1$.
- Example. If $f_X(x) = \frac{1}{\pi(1+x^2)}, x \in \mathbb{R}$, $\int_{-\infty}^{\infty} |x| f_X(x) dx = 2\int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \frac{1}{\pi} \ln(1+x^2)|_0^{\infty} = \infty$

Expectation of Random Variables

- Let $Y = r(X)$. Then $E(Y) = E\big(r(X)\big) = \int r(x) dF_X(x)$.
- Let A be an event and let $r(x) = I_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$. Then

$$E\big(I_A(X)\big) = \int I_A(x)\, dF_X(x) = \int_A dF_X(x) = P(X \in A).$$

- Let $X \sim \text{Uniform}(0,1)$. Let $Y = e^X$. Then $E(Y) = \int_0^1 e^x dx = e - 1$. Since $x = \ln y$ and $\frac{dx}{dy} = \frac{1}{y}$, $f_Y(y) = \frac{1}{y} I(1 < y < e)$. $E(Y) = \int_1^e y f_Y(y) dy = e - 1$.
- Take a stick of unit length and break it at random. Let Y be the length of the longer piece. What is the mean of Y?
- If X is the break point, then $X \sim \text{Uniform}(0,1)$ and $Y = \max(X, 1 - X)$.

$$E(Y) = \int_0^1 \max(x, 1-x)\, dx = \int_0^{0.5} (1-x) dx + \int_{0.5}^1 x dx = \frac{3}{4}$$

Expectation of Random Variables

- If $Z = r(X,Y)$ then $E(Z) = \iint r(x,y)dF_{(X,Y)}(x,y)$.
- Example. Let (X,Y) have a jointly uniform distribution on the unit square. Let $Z = X^2 + Y^2$. Then,

$$E(Z) = \int_0^1 \int_0^1 (x^2 + y^2)dxdy = \int_0^1 x^2 dx + \int_0^1 y^2 dy = \frac{2}{3}.$$

- The kth moment of X is defined to be $E(X^k)$ assuming that $E(|X|^k) < \infty$.
- If the kth moment exists, and if $j < k$ then the jth moment exists.
- The kth central moment is defined to be $E\left((X - E(X))^k\right)$.

$$\int |x|^j dF_X(x) = \int_{|x|<1} |x|^j dF_X(x) + \int_{|x|>1} |x|^j dF_X(x)$$

$$\leq \int_{|x|<1} dF_X(x) + \int_{|x|>1} |x|^k dF_X(x) \leq 1 + E(|X|^k) < \infty$$

Expectation of Random Variables

- If $X_1, X_2, \ldots, X_n$ are random variables and $a_1, a_2, \ldots, a_n$ are constants, then

$$E\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i E(X_i)$$

- Example. Let $X \sim \text{Binomial}(n, p)$. What is the mean of X?
  - Since $X = \sum_{j=1}^{n} X_j$ where $X_j \sim \text{Bernoulli}(p)$ and $X_1, X_2, \ldots, X_n$ are independent,

$$E(X) = \sum_{j=1}^{n} E(X_j) = np.$$

- Let $X_1, X_2, \ldots, X_n$ be independent random variables. Then, $E\left(\prod_{j=1}^{n} X_j\right) = \prod_{j=1}^{n} E(X_j)$

Variance and Covariance

- Let X be a random variable with mean μ. The variance of X, denoted by σ_X^2 or $Var(X)$, is defined by

$$\sigma_X^2 = E\left((X-\mu)^2\right)$$

  assuming the expectation exists. The standard deviation is $\sigma_X = \sqrt{\sigma_X^2}$.

- Assuming the variance is well-defined, it has the following properties:
  - $\sigma_X^2 = E(X^2) - \mu^2$
  - If a and b are constants, then $Var(aX+b) = a^2 Var(X)$
  - If $X_1, X_2, \ldots, X_n$ are independent and $a_1, a_2, \ldots, a_n$ are constants, then

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i)$$

- Example. Let $X \sim$ Binomial(n,p). Since $X = \sum_{j=1}^{n} X_j$ where $X_j \sim$ Bernoulli(p) and $X_1, X_2, \ldots, X_n$ are independent, $Var(X) = \sum_{j=1}^{n} Var\left(X_j\right) = np(1-p)$. Note that $E\left(X_j^2\right) = E\left(X_j\right) = p$.

Variance and Covariance

- If $X_1, X_2, \ldots, X_n$ are random variables with mean μ and variance σ^2 then we define the sample mean to be $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j$ and the sample variance to be $S_n^2 = \frac{1}{n-1}\sum_{j=1}^{n}\left(X_j - \bar{X}_n\right)^2$.
- Note that $E(\bar{X}_n) = \frac{1}{n} \cdot n\mu = \mu$.
- Note that

$E(X_i - \bar{X}_n) = 0$

$\{X_1, \ldots, X_n\}$ are independent

$$E\left((X_i - \bar{X}_n)^2\right) = Var(X_i - \bar{X}_n) = Var\left(X_i - \frac{1}{n}\sum_{j=1}^{n} X_j\right) = Var\left(\left(1 - \frac{1}{n}\right)X_i - \frac{1}{n}\sum_{j\neq i} X_j\right)$$

$$= \left(1 - \frac{1}{n}\right)^2 \sigma^2 + \frac{n-1}{n^2}\sigma^2 = \frac{n-1}{n}\sigma^2 \Rightarrow E(S_n^2) = \frac{1}{n-1} \cdot \frac{n-1}{n}\sigma^2 = \sigma^2.$$

Variance and Covariance

- Let X and Y be random variables with means μ_X and μ_Y and standard deviations σ_X and σ_Y. Define the covariance between X and Y by $Cov(X,Y) = E\big((X-\mu_X)(Y-\mu_Y)\big)$ and the correlation by $\rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$.
- The covariance satisfies $Cov(X,Y) = E(XY) - E(X)E(Y)$.
- The correlation satisfies $-1 \le \rho_{X,Y} \le 1$.

  For $t \in \mathbb{R}$, $E\left[\big((Y-\mu_Y) - t(X-\mu_X)\big)^2\right] \ge 0 \Longrightarrow \sigma_Y^2 - 2t\rho_{X,Y}\sigma_X\sigma_Y + t^2\sigma_X^2 \ge 0 \Longrightarrow Discriminant \le 0$

  $\Longrightarrow 4\rho_{X,Y}^2\sigma_X^2\sigma_Y^2 - 4\sigma_X^2\sigma_Y^2 \le 0 \Longrightarrow \rho_{X,Y}^2 \le 1.$

- If $Y = aX + b$ for some constants a and b then $\rho_{X,Y} = 1$ if $a > 0$ and $\rho_{X,Y} = -1$ if $a < 0$.
- If X and Y are independent then $Cov(X,Y) = 0$. But the converse is not true.
- $Var(X \pm Y) = Var(X) + Var(Y) \pm 2\,Cov(X,Y)$. For random variables $X_1, X_2, \ldots, X_n$,

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} a_i a_j Cov\big(X_i, X_j\big).$$

Variance and Covariance

- Let $X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_k \end{pmatrix}$ be a random vector.
- $E(X) = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_k \end{pmatrix} = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_k) \end{pmatrix}$ and $Var(X) = \begin{pmatrix} Var(X_1) & Cov(X_1, X_2) & \cdots & Cov(X_1, X_k) \\ Cov(X_2, X_1) & Var(X_2) & \cdots & Cov(X_2, X_k) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_k, X_1) & Cov(X_k, X_2) & \cdots & Var(X_k) \end{pmatrix}$

Example. If $X \sim$ Multinomial(n, p) then $E(X) = n(p_1, p_2, \ldots, p_k)$ and

$$Var(X) = \begin{pmatrix} np_1(1-p_1) & -np_1p_2 & \cdots & -np_1p_k \\ -np_2p_1 & np_2(1-p_2) & \cdots & -np_2p_k \\ \vdots & \vdots & \ddots & \vdots \\ -np_kp_1 & -np_kp_2 & \cdots & np_k(1-p_k) \end{pmatrix}$$

Variance and Covariance

- Note that $X_i \sim$Binomial(n, p_i) and $X_i + X_j \sim$Binomial$\left(n, p_i + p_j\right)$
- $E(X_i) = np_i$ and $Var(X_i) = np_i(1 - p_i)$
- $E\left(X_i + X_j\right) = n\left(p_i + p_j\right)$ and $Var\left(X_i + X_j\right) = n\left(p_i + p_j\right)\left(1 - \left(p_i + p_j\right)\right)$
- Since $Var\left(X_i + X_j\right) = Var(X_i) + Var\left(X_j\right) + 2Cov\left(X_i, X_j\right)$,

$$2Cov\left(X_i, X_j\right) = n\left(p_i + p_j\right)\left(1 - \left(p_i + p_j\right)\right) - np_i(1 - p_i) - np_j\left(1 - p_j\right) = -2np_ip_j$$

$$\Longrightarrow Cov\left(X_i, X_j\right) = -np_ip_j.$$

- If A is a matrix of constants and X is a random vector with mean μ and variance Σ, then
  - $E(AX) = A\mu$
  - $Var(AX) = A\Sigma A'$

Variance and Covariance

- If $X_1, X_2, \ldots, X_n$ are random variables then we define the sample mean to be $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j$ and the sample variance to be $S_n^2 = \frac{1}{n-1}\sum_{j=1}^{n}\left(X_j - \bar{X}_n\right)^2$.

$$\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j = \frac{1}{n}\begin{pmatrix}1 & 1 & \cdots & 1\end{pmatrix}\begin{pmatrix}X_1\\X_2\\\vdots\\X_n\end{pmatrix} = \frac{1}{n}1_n'X = \frac{1}{n}X'1_n$$

$$S_n^2 = \frac{1}{n-1}\sum_{j=1}^{n}\left(X_j - \bar{X}_n\right)^2 = \frac{1}{n-1}\left[\sum_{j=1}^{n} X_j^2 - n\bar{X}_n^2\right] = \frac{1}{n-1}\left[X'X - n\frac{1}{n}X'1_n\frac{1}{n}1_n'X\right]$$

$$= \frac{1}{n-1}X'\left(I_n - \frac{1}{n}1_n1_n'\right)X$$

Variance and Covariance

- Let $X_1, X_2, \ldots, X_n$ be IID and let $\mu = E(X_i)$, $\sigma^2 = Var(X_i)$. Then, $E(\bar{X}_n) = \mu$, $Var(\bar{X}_n) = \frac{\sigma^2}{n}$ and $E(S_n^2) = \sigma^2$.
- Note that $E(X) = \mu 1_n$. $E(\bar{X}_n) = E\left(\frac{1}{n} 1_n' X\right) = \frac{1}{n} 1_n' \mu 1_n = \mu$.
- Note that $Var(X) = \sigma^2 I_n$. $Var(\bar{X}_n) = Var\left(\frac{1}{n} 1_n' X\right) = \frac{1}{n^2} 1_n' Var(X) 1_n = \frac{\sigma^2}{n}$.

Trace of a scalar is equal to the scalar.

$$E(S_n^2) = E\left(\frac{1}{n-1} X'\left(I_n - \frac{1}{n} 1_n 1_n'\right) X\right) = \frac{1}{n-1} E\left(tr\left(X'\left(I_n - \frac{1}{n} 1_n 1_n'\right) X\right)\right)$$

$$= \frac{1}{n-1} E\left(tr\left(\left(I_n - \frac{1}{n} 1_n 1_n'\right) XX'\right)\right) = \frac{1}{n-1} tr\left(\left(I_n - \frac{1}{n} 1_n 1_n'\right)(\sigma^2 I_n + \mu^2 1_n 1_n')\right)$$

$$= \frac{\sigma^2}{n-1} tr\left(I_n - \frac{1}{n} 1_n 1_n'\right) = \sigma^2$$

tr(AB) = tr(BA)

E(tr(AX)) = tr(E(AX)) = tr(AE(X))

Conditional Expectation

- The conditional expectation of X given $Y = y$ is

$$E(X|Y=y) = \begin{cases} \sum_x xP(X=x|Y=y) & \text{discrete case} \\ \int_{-\infty}^{\infty} x f_{X|Y}(x|y)dx & \text{continuous case} \end{cases}$$

- If $r(x,y)$ is a function of x and y then

$$E(r(X,Y)|Y=y) = \begin{cases} \sum_x r(x,y)P(X=x|Y=y) & \text{discrete case} \\ \int_{-\infty}^{\infty} r(x,y) f_{X|Y}(x|y)dx & \text{continuous case} \end{cases}$$

Conditional Expectation

- Example. Suppose we draw $X \sim \text{Uniform}(0,1)$ and $Y|X = x \sim \text{Uniform}(x, 1)$.
- Since $E(Y|X = x) = \frac{1+x}{2}$, $E(Y|X) = \frac{1+X}{2}$.

  Since $f_{Y|X}(y|x) = \frac{1}{1-x} I(x < y < 1)$,

$$E(Y|X = x) = \int_x^1 \frac{y}{1-x} dy = \frac{1+x}{2}$$

- The Rule of Iterated Expectations. For random variables X and Y, assuming the expectations exist, we have that $E\big(E(Y|X)\big) = E(Y)$ and $E\big(E(X|Y)\big) = E(X)$.
- For any function $r(x, y)$, we have $E\big(E(r(X,Y)|X)\big) = E\big(r(X,Y)\big)$

$$E\big(E(Y|X)\big) = \int E(Y|X = x) dF_X(x) = \int\int y f_{Y|X}(y|x) f_X(x)\, dy dx = \int\int y f_{X,Y}(x,y) dx dy = \int y f_Y(y) dy$$

$$= E(Y).$$

- Example (Continuous) Since $E(Y|X) = \frac{1+X}{2}$, $E(Y) = E\big(E(Y|X)\big) = \frac{1+E(X)}{2} = \frac{3}{4}$.

Conditional Expectation

- The conditional variance is defined as

$$Var(Y|X=x) = \int \big(y - E(Y|X=x)\big)^2 f_{Y|X}(y|x)dy.$$

- For random variables X and Y, $Var(Y) = E\big(Var(Y|X)\big) + Var\big(E(Y|X)\big)$.

Since $Var(X|Y=y) = E(X^2|Y=y) - E(X|Y=y)^2$,

$$E\big(Var(X|Y)\big) = E\big(E(X^2|Y)\big) - E(E(X|Y)^2) = E(X^2) - \Big(Var\big(E(X|Y)\big) + E\big(E(X|Y)\big)^2\Big)$$
$$= (E(X^2) - E(X)^2) - Var\big(E(X|Y)\big) = Var(X) - Var\big(E(X|Y)\big)$$

Conditional Expectation

- Example. Suppose $Q \sim \text{Uniform}(0,1)$ and $X|Q = q \sim \text{Binomial}(n, q)$.

$$E(X) = E\big(E(X|Q)\big) = E(nQ) = \frac{n}{2}.$$

$$E(Q^2) = \int_0^1 x^2 dx = \frac{1}{3};\ Var(Q) = E(Q^2) - E(Q)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$Var(X) = Var\big(E(X|Q)\big) + E\big(Var(X|Q)\big) = Var(nQ) + E\big(nQ(1-Q)\big) = \frac{n^2}{12} + n\left(\frac{1}{2} - \frac{1}{3}\right)$$

$$= \frac{n}{6} + \frac{n^2}{12}$$

Conditional Independence

- Let X, Y and Z be random variables. X and Y are conditionally independent given Z, written $X \perp Y|Z$, if $f_{X,Y|Z}(x,y|z) = f_{X|Z}(x|z)f_{Y|Z}(y|z)$ for all x, y and z.
  - Intuitively, this means that, once we know Z, Y provides no extra information about X.
  - $f_{X|Y,Z}(x|y,z) = f_{X|Z}(x|z)$

$$f_{X|Y,Z}(x|y,z) = \frac{f_{X,Y|Z}(x,y|z)}{f_{Y|Z}(y|z)} = \frac{f_{X|Z}(x|z)f_{Y|Z}(y|z)}{f_{Y|Z}(y|z)} = f_{X|Z}(x|z)$$

  - $X \perp Y|Z \Rightarrow Y \perp X|Z$
  - $X \perp Y|Z$ and $U = h(X) \Longrightarrow U \perp Y|Z$
  - $X \perp Y|Z$ and $U = h(X) \Longrightarrow X \perp Y|(Z,U)$

$$f_{X|Y,Z,U}(x|y,z,u) \propto f_{X|Y,Z}(x|y,z)I(h(x) = u)$$
$$= f_{X|Z}(x|z)I(h(x) = u) \Longrightarrow f_{X|Y,Z,U}(x|y,z,u) = f_{X|Z,U}(x|z,u)$$

Conditional Independence

- $X \perp Y|Z$ and $X \perp W|(Y,Z) \Longrightarrow X \perp (W,Y)|Z$

$$X \perp Y|Z \Longrightarrow f_{X|Y,Z}(x|y,z) = f_{X|Z}(x|z)$$
$$X \perp W|(Y,Z) \Longrightarrow f_{X|W,Y,Z}(x|w,y,z) = f_{X|Y,Z}(x|y,z) = f_{X|Z}(x|z)$$

- $X \perp Y|Z$ and $X \perp Z|Y \Longrightarrow X \perp (Y,Z)$

$X \perp Y|Z \Longrightarrow f_{X|Y,Z}(x|y,z) = f_{X|Z}(x|z)$ depending on z only.
$X \perp Z|Y \Longrightarrow f_{X|Y,Z}(x|y,z) = f_{X|Y}(x|y)$ depending on y only.

This can only be true for all y and z when $f_{X|Z}(x|z) = f_{X|Y}(x|y) = f_X(x)$

Conditional Independence

- Let Z be a random variable.
- Suppose $X = Z + e_1$ and $Y = Z + e_2$ where
  - $e_1, e_2 \sim n(0,1)$
  - Z, e_1, e_2 are independent.
- Unconditionally, $Cov(X, Y) = Cov(Z + e_1, Z + e_2) = Var(Z)$
- $Cov(X, Y|Z) = Cov(e_1, e_2) = 0$.